PAEDIATRIC ORTHO · NEET-SS / FET / INI-SS

Perthes and SCFE

Part A: Legg-Calvé-Perthes disease (LCPD)

1. Definition and epidemiology

- **Idiopathic avascular necrosis of the capital femoral epiphysis in a child**, followed by **fragmentation, revascularisation and remodelling** (self-limiting but may leave a deformed head).
- **Age 4-8 years (range 2-12)**; **boys about 4-5:1**; **bilateral in about 10-15%** (**asynchronous**: **symmetric simultaneous bilateral involvement suggests an alternative diagnosis: multiple epiphyseal dysplasia (MED), hypothyroidism, sickle cell, Gaucher, spondyloepiphyseal dysplasia**).
- **Associations: delayed bone age (about 90%)**, **short stature**, **low birth weight**, **ADHD/hyperactivity**, **passive smoking**, **low socioeconomic status**, **thrombophilia (protein C/S deficiency, factor V Leiden: controversial)**, **inguinal hernia**, genitourinary anomalies.

2. Pathogenesis

- **Single or repeated interruption of the blood supply** (**lateral epiphyseal vessels from the medial femoral circumflex artery**) to the capital femoral epiphysis (**at age 4-7, the epiphysis depends almost entirely on the lateral epiphyseal vessels**: the **ligamentum teres artery contributes little** and the **metaphyseal vessels do not cross the physis**).
- **Stages (Waldenström):**
  - **Initial (sclerosis/necrosis)**: **sclerotic, smaller ossific nucleus**, **widened medial joint space**, **subchondral fracture ("crescent**

sign": Caffey sign): about 6 months,
- **Fragmentation (resorption)**: **fragmented epiphysis**, lateral extrusion, **most mechanically vulnerable stage** (deformation): about 1 year,
- **Reossification (healing)**: new bone formation: about 3-5 years,
- **Residual (healed)**: final shape (**coxa magna, coxa plana, short neck, trochanteric overgrowth**).

3. Clinical features

- **Painless (or mild) limp** (**commonest presentation**), **pain referred to the knee/thigh** (obturator nerve), **antalgic or Trendelenburg gait**, **limited abduction and internal rotation** (most important; **loss of abduction = poor prognosis**), thigh atrophy, mild shortening.
- **Differential diagnosis: transient synovitis**, **septic arthritis**, **MED/SED**, **hypothyroidism**, **sickle cell disease**, **Gaucher disease**, **steroid-induced AVN**, **Meyer dysplasia** (**irregular ossification of the femoral head under 4 years**, often bilateral: **self-resolving**).

4. Imaging

- **Radiographs (AP and frog-leg lateral)**: stages, **crescent sign (subchondral fracture: earliest radiographic sign)**, head-at-risk signs.
- **MRI** (early diagnosis, extent of necrosis, **perfusion MRI**), **bone scan** (avascularity; early).
- **Arthrography** (dynamic: **hinge abduction**, containability).

5. Classifications

Herring lateral pillar classification (**most widely used; prognosis; assessed at the start of the fragmentation stage**)

- **Lateral pillar** = lateral 15-30% of the epiphysis on AP radiograph.

Group	Lateral pillar height	Prognosis
A	**Intact (100%)**	**Good regardless of age**
B	**Over 50% maintained**	Good if **under 8 years at onset** (bone age under 6)
B/C border	**About 50%**, narrow (2-3 mm) or poorly ossified pillar, or exactly 50%	Intermediate
C	**Under 50% maintained**	**Poor** (regardless of treatment, especially older)

Catterall classification (extent of head involvement: I-IV; interobserver reliability poor)

- **I: anterior head only (under 25%)**; **II: under 50% (anterior + central sequestrum)**; **III: over 50% (only medial/posterior spared)**; **IV: whole head**.
- **Catterall "head-at-risk" signs (poor prognosis):**
  - **Gage sign** (V-shaped radiolucency in the lateral epiphysis/metaphysis),
  - **Calcification lateral to the epiphysis** (lateral extrusion),
  - **Lateral subluxation of the femoral head**,
  - **Horizontal growth plate**,
  - **Metaphyseal cysts (diffuse metaphyseal reaction)**.

Salter-Thompson classification

- **A: subchondral fracture under 50% of the head; B: over 50%** (prognostic; in the early stage).

Stulberg classification (outcome at maturity: predicts OA risk)

Class	Head shape	Congruency	OA risk
I	**Normal (spherical)**	**Spherical congruency**	**None**

Class	Head shape	Congruency	OA risk
II	**Spherical but larger (coxa magna), short neck or steep acetabulum**	**Spherical congruency**	Minimal
III	**Ovoid/mushroom-shaped (not flat)**	**Aspherical congruency**	**Mild-moderate OA in later adulthood**
IV	**Flat head (over 1 cm flat weight-bearing surface) + flat acetabulum**	**Aspherical congruency**	**Moderate OA (in middle age)**
V	**Flat head with normal acetabulum**	**Aspherical incongruency**	**Severe early OA**

6. Prognostic factors

- **Age at onset (most important): under 6 years: good prognosis** (more remodelling time); **over 8 years: poor**.
- **Lateral pillar classification (Herring)** (most important radiographic predictor), **extent of head involvement**, **head-at-risk signs**, **female sex** (worse: earlier skeletal maturity), **loss of hip motion (abduction)**, **hinge abduction**, **Stulberg class at maturity**.

7. Management (principle: containment)

- **Goals: maintain the femoral head within the acetabulum (containment)** during the vulnerable fragmentation stage, so that the acetabulum acts as a mould; **maintain ROM** (especially **abduction**).
- **Non-operative:**
  - **Under 6 years (bone age), all groups (mostly): observation, activity modification, physiotherapy (ROM, abduction stretching)**, NSAIDs for synovitis; **bracing (Petrie**

casts/abduction orthoses): largely abandoned (no proven benefit).

- **Operative containment (Herring prospective study: 2004):**
  - **Over 8 years (chronological; bone age over 6) with lateral pillar B or B/C**: **surgical containment (proximal femoral varus osteotomy or Salter innominate osteotomy) gives better results than non-operative treatment**.
  - **Group A**: good outcome regardless.
  - **Group C**: poor outcome regardless of treatment (in most series).
  - **Under 8 years with B**: good outcomes without surgery.
- **Containment procedures:**
  - **Proximal femoral varus (derotation) osteotomy (VDRO)**: **about 15-20 degrees varus**; **shortening, Trendelenburg gait (abductor weakness), trochanteric overgrowth** (often remodels in young children),
  - **Salter innominate osteotomy** (**anterolateral coverage**; lengthens slightly; **requires near-full abduction**),
  - **Shelf acetabuloplasty**, **triple innominate osteotomy**, **combined femoral and pelvic osteotomies** for severe cases.
- **Hinge abduction (salvage: non-containable head; the extruded head hinges on the lateral acetabular edge in abduction):**
  - **Valgus (extension) osteotomy** (**brings the congruent part of the head into the weight-bearing zone; improves adduction-abduction mechanics**), **Chiari osteotomy**, **shelf procedure**, **cheilectomy** (removing the lateral prominence: risk of AVN/physeal injury).
- **Late sequelae:**
  - **Trochanteric overgrowth (relative)**: **greater trochanteric epiphysiodesis (under 8 years)** or **trochanteric advancement (Wagner)**.

- **FAI (cam-type from a mushroom head)**: **surgical hip dislocation with femoral head reduction osteotomy**, **relative femoral neck lengthening**, osteochondroplasty.
- **Osteochondritis dissecans of the femoral head**, **labral tears**, **OA (THA in adulthood)**.

Part B: Slipped capital femoral epiphysis (SCFE)

8. Definition and epidemiology

- **Displacement of the proximal femoral metaphysis anteriorly and superiorly (externally rotated) relative to the epiphysis** (the **epiphysis stays in the acetabulum** and is **posterior and inferior** relative to the neck) through the **hypertrophic zone of the physis**.
- **Commonest adolescent hip disorder**: **age 10-16 (boys: about 13-14; girls: about 11-12)**, **boys more than girls (about 1.5-2:1)**, **left more than right**, **bilateral in about 20-40% (up to 60% eventually; about 20% at presentation)**.
- **Obesity (strongest association: over 95th percentile BMI in about 50-80%)**, **African-American and Polynesian ethnicity**.
- **Endocrine/metabolic causes (atypical SCFE: suspect if under 10 years or over 16, below 50th percentile weight, short stature): hypothyroidism (commonest endocrine)**, **growth hormone deficiency/supplementation**, **renal osteodystrophy (chronic renal failure)**, **hypogonadism**, **panhypopituitarism**, **radiotherapy**, **Down syndrome**. **Bilateral slips are much more common in endocrine SCFE** (**prophylactic pinning recommended**).
- **Biomechanics: increased shear forces** (obesity, **femoral retroversion**, **physeal obliquity: more vertical physis in adolescence**), **perichondrial ring thinning** during growth spurt, **hormonal (oestrogen protects; testosterone weakens the physis)**.

9. Classification

Loder (stability; prognostic: **the most important**)

- **Stable: able to bear weight (with or without crutches)**: **AVN risk about 0-5%**.
- **Unstable: unable to bear weight (even with crutches)**: **AVN risk about 20-50%** (Loder: 47%).

Temporal (traditional)

- **Acute (symptoms under 3 weeks)**, **chronic (over 3 weeks: commonest; remodelling of the metaphysis, posteroinferior callus)**, **acute-on-chronic**.

Southwick slip angle (severity; on frog-leg lateral)

- **Epiphyseal-shaft angle difference vs the contralateral normal side: mild under 30 degrees**, **moderate 30-50 degrees**, **severe over 50 degrees**.

Wilson (displacement on AP)

- **Mild under 33%**, **moderate 33-50%**, **severe over 50%** of the metaphyseal width.

10. Clinical features

- **Groin, thigh or KNEE pain** (**obturator nerve referred pain: knee pain in an adolescent = examine the hip**: **delayed diagnosis common**), **limp**, **out-toeing**.
- **Obligatory external rotation with hip flexion (Drehmann sign)**, **loss of internal rotation** (most sensitive), loss of abduction and flexion, **shortening**.
- **Unstable:** acute severe pain, **unable to bear weight**, limb externally rotated and shortened (like a femoral neck fracture).

11. Imaging

- **AP and frog-leg lateral pelvis** (**frog-leg lateral avoided in unstable slips**: **cross-table lateral instead**: may displace further).
- **Signs:**
  - **Klein line** (**line along the superior femoral neck on AP should intersect the lateral epiphysis; in SCFE it does not or intersects less than the normal side**: **Trethowan sign**),
  - **Widened, irregular physis** (early pre-slip),
  - **Metaphyseal blanch sign of Steel** (**double density: overlap of the displaced epiphysis on the metaphysis**),
  - **Reduced epiphyseal height**,
  - **Capener sign** (loss of the normal triangular overlap of the medial metaphysis with the posterior acetabular wall).
- **MRI: pre-slip (physeal widening, marrow oedema)**; the contralateral hip; AVN.
- **Bone age:** Oxford, **triradiate cartilage open = higher contralateral slip risk**; **Risser**; **modified Oxford score**.

12. Management

Stable SCFE

- **Urgent (not emergency) admission, non-weight bearing**, then **in situ fixation with a single cannulated screw** (**"pin in situ": do NOT attempt reduction: AVN risk**):
  - **Screw perpendicular to the physis, in the centre of the epiphysis on both views**, entering **anterior on the femoral neck** (because the epiphysis is posterior: **anterior-to-posterior trajectory**),
  - **At least 5 threads (about 4-5 threads) across the physis**; **tip at least 5 mm from the subchondral bone** (**avoid joint**

penetration: chondrolysis: **"approach-withdraw" technique** and multiple fluoroscopic views to exclude penetration),
    - **Single central screw for stable slips** (**two screws for unstable slips** add rotational stability but increase penetration risk).
- **Physeal closure** after fixation usually within about 1 year.

Unstable SCFE

- **Emergency (controversial timing): gentle positional/incidental reduction**, **capsular decompression (arthrotomy/aspiration: reduces intracapsular tamponade)**, **fixation with one or two screws**; **urgent fixation within 24 hours** (some evidence for lower AVN); **avoid forceful manipulation**.
- **Modified Dunn procedure (subcapital realignment via surgical hip dislocation: Ganz approach)**: **anatomic reduction with preservation of the retinacular vessels** (**extended retinacular flap**); for **unstable and severe stable slips** in experienced centres; **AVN about 0-25% (debated)**; **intraoperative monitoring of head perfusion (drill hole bleeding, ICP monitoring)**.

Severe chronic/stable slips with deformity

- **In situ pinning followed by corrective osteotomy for symptomatic FAI/deformity later:**
    - **Subcapital (cuneiform: Fish/Dunn)** (**closest to the deformity: best correction, highest AVN**),
    - **Base of neck (Kramer, Barmada)** (intermediate),
    - **Intertrochanteric (Imhäuser, Southwick: flexion-valgus-internal rotation osteotomy)** (**safest for the blood supply, compensatory, less correction**),
    - **Arthroscopic/open osteochondroplasty** for cam deformity.

Prophylactic pinning of the contralateral hip

- **Recommended for: endocrine/renal SCFE**, **age under 10 years**, **open triradiate cartilage**, **poor follow-up compliance**, **modified**

Oxford bone age score under 16-19, **posterior sloping angle (PSA) over about 12-14 degrees** (**increased risk of contralateral slip**). Otherwise: close follow-up.

13. Complications

- **AVN** (**unstable slips (up to about 50%)**, **forceful reduction**, **superior/posterosuperior screw placement** (lateral epiphyseal vessels), **subcapital osteotomy**): **the most serious complication**.
- **Chondrolysis** (**acute cartilage necrosis: joint space narrowing to under 3 mm**, painful stiffness): **unrecognised screw penetration (commonest cause)**, spica casting, **African-American/female**; **ROM physio, NSAIDs**; may need arthrodesis/THA.
- **Slip progression** (fixation failure), **contralateral slip**, **FAI (cam; pistol grip)**, **OA**, **leg-length discrepancy**, **subtrochanteric fracture** (**entry at or below the lesser trochanter** with multiple drill holes: stress riser), infection.

Exam pearl: **Perthes: 4-8-year-old boys, delayed bone age, painless limp, loss of abduction/IR; bilateral symmetrical = think MED/hypothyroidism. Waldenström stages; crescent sign earliest. Herring lateral pillar (A/B/B-C/C) assessed in fragmentation; Catterall head-at-risk signs (Gage, lateral calcification, subluxation, horizontal physis, metaphyseal cysts).**

Exam pearl: **Age at onset is the key prognostic factor (under 6 good). Over 8 years with B or B/C: surgical containment (femoral varus or Salter osteotomy). Hinge abduction: valgus osteotomy. Stulberg I-II good, III-IV moderate, V severe early OA (aspherical incongruity).**

Exam pearl: **SCFE: obese adolescent with knee pain; Klein line, Steel blanch sign; Loder stable (AVN under 5%) vs unstable (up to 50%). Pin in situ with one central screw perpendicular to the physis, no reduction; joint penetration = chondrolysis. Atypical (under 10, thin, endocrine/renal) = bilateral: pin the other side. Imhäuser (intertrochanteric) safest osteotomy; modified Dunn for severe/unstable in expert hands.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.